

1. Basics of Computers (Computer ki Buniyad)

1.1 What is a Computer? (Computer Kya Hai?)

Simple Definition: Computer ek **electronic machine** hai jo **data** leta hai (Input), us data ko **process** karta hai (Processing), aur hamein **useful information** deta hai (Output). Ye data ko future ke liye **store** bhi kar sakta hai.

Agar hum isko ek **real-life example** se samjhein:

Aapke paas ek calculator hai. Aap usmein **5** aur **3** type karte hain (ye hai **Input**). Calculator andar se calculate karta hai (ye hai **Processing**) aur screen pe result **8** dikhata hai (ye hai **Output**). Computer bhi exactly yahi kaam karta hai, bas bahut zyada advanced tareeke se aur bahut saari cheezein ek saath karta hai.

Naya Term Samjhiye:

- **Data:** Kachchi information, jiska koi matlab nahi bana hai. Jaise, **1998** , **Shivani** , **95%** . Ye alag-alag facts hain.
- **Information:** Jab data ko process karke meaningful bana diya jaye. Jaise, "Shivani ka janam 1998 mein hua aur uske 12th mein 95% marks aaye." Ye information hai.

1.2 Characteristics of Computers (Computer ki Khasiyat)

Computer mein kuch amazing qualities hoti hain jo use insaan se alag banati hain:

1. **Speed (Gati):** Computer bahut tez kaam karta hai. Woh ek second mein lakhon calculations kar sakta hai. Uski speed **Megahertz (MHz)** ya **Gigahertz (GHz)** mein maapi jaati hai.
 - *Analogy:* Jaise ek super-fast bike 1 minute mein 2 kilometer chali jaye, waise hi computer 1 second mein karodon instructions process kar leta hai.
2. **Accuracy (Shuddhata):** Computer 100% accurate kaam karta hai. Agar galti hoti hai, to woh insaan ki di hui galat input ya program ki wajah se hoti hai, computer ki apni galati nahi hoti. Is phenomenon ko **GIGO (Garbage In, Garbage Out)** kehte hain.
 - *Memory Trick:* GIGO yaad rakho — agar aap kachra (galat data) doge,

to computer bhi kachra (galat result) hi dega.

3. **Diligence (Thakaan se Pare):** Computer kabhi thakta nahi hai. Woh bina ruke, bina speed kam kiye, ek jaise consistent tarah se lagatar ghanton kaam kar sakta hai.
 - *Analogy:* Insaan 2 ghante padhke thak jaata hai, computer 2 mahine bina roke kaam karega, uski speed same rahegi.
4. **Storage (Bhandaran Kshamta):** Computer data ko store kar sakta hai. Uski memory bahut badi hoti hai. Aap lakho books ki information ek choti si hard disk mein rakh sakte hain.
 - *Types:* Temporary storage (RAM - jaise aapki padhte waqt ki yaaddaasht) aur Permanent storage (HDD/SSD - jaise aapki kitaab).
5. **Versatility (Bahumukhi Prativa):** Ek hi computer se aap kaafi alag-alag kaam kar sakte hain — kabhi aap uspe film dekh rahe hain, kabhi document type kar rahe hain, kabhi game khel rahe hain, kabhi internet chala rahe hain. Yehi versatility hai.
6. **Automation (Swayam Kriya):** Ek baar program de diya, to computer khud automatically kaam karta hai, baar-baar instruction dene ki zaroorat nahi hai.
 - *Example:* Aapne alarm set kiya subah 6 baje ka. Alarm apne aap bajega, aapko roz bolna nahi padega.
7. **No IQ (Buddhi Ka Abhav):** Computer ke paas apni koi akal ya sochne ki shakti nahi hai. Woh sirf wahi karega jo aap use karne ko kahoge. Artificial Intelligence aaj kal isko badal rahi hai, but fundamentally, computer ko apni buddhi nahi hoti.
 - *Exam Point:* Computer has IQ zero. It works on stored instructions.

1.3 History of Computers (Computer ka Itihaas)

Computer kisi ek din mein nahi ban gaya. Iske peeche hazaaron saalon ki mehnat aur kaafi saare tools ka evolution hai.

Tool/Device	Approx Year	Use / Significance
Abacus	~2400 BC	Pehla calculating tool. Isme beads (daane) hote the jo rods par slide karte the. (Jodne-ghatane ke liye)

Pascaline	1642	Blaise Pascal ne banayi. Pehla mechanical calculator. Sirf jod aur ghatana kar sakta tha.
Step Reckoner	1673	Leibniz ne banaya. Multiply and divide bhi kar sakta tha.
Difference Engine	1822	Charles Babbage ne design kiya. Steam powered, complex mathematical tables banane ke liye.
Analytical Engine	1837	Charles Babbage ka sabse bada contribution. Isme Input, Output, Memory, aur CPU jaisi sabhi modern computer ki khasiyatein thin. Isliye Charles Babbage ko "Father of the Computer" kehte hain.
Ada Lovelace	1840s	Ada Lovelace ne Analytical Engine ke liye pehla program likha. Isliye unhe duniya ki "First Programmer" kaha jaata hai.
Tabulating Machine	1890	Herman Hollerith ne banayi. Census (janganana) ke data ko punch cards se process karne ke liye. Hollerith ki company baad mein IBM ban gayi.

1.4 Generations of Computers (Computer ki Peedhiyaan)

Jab technology change hui, to computer ki generation badalti gayi. Har generation mein size, speed, aur reliability mein improvement aaya.

Generation	Period	Key Technology	Key Features
First	1940-1956	Vacuum Tubes	Bohot bade (room-size), bahut garam, mahange, zyada bijli khane wale. Machine language use hoti thi. Example: ENIAC, UNIVAC
Second	1956-1962	Transistors	Vacuum tubes se chhote, saste, kam garam, tez. Assembly language aayi.

	1963		Example: IBM 1620, IBM 7094
Third	1964-1971	Integrated Circuits (ICs)	Ek chip mein kai transistors. Aur chhote, aur bharosemand. Keyboard, monitor aaye. Operating system ka concept aaya. Example: IBM-360 series
Fourth	1971-1980	VLSI / Microprocessor	Ek chip mein lakhon transistors . Isse Microprocessor bana. Personal Computer (PC) ka janam hua. Aaj bhi hum isi generation ko modern maante hain. GUI, Mouse, Laptop aaye. Example: Intel 4004, IBM PC
Fifth	1980-Present	ULSI / AI	Ek chip mein karodon transistors . Artificial Intelligence, parallel processing, quantum computing, cloud computing par focus. Natural language, speech recognition.

Memory Trick for Generations: "Very Tall Indian Visitors Use" -> Vacuum Tube, Transistors, Integrated Circuits, VLSI/ MicroProcessor, ULSI / AI. (P ko V bolke adjust karo, ya "Very Talented Indian Players Use" AI kar lo).

1.5 Types of Computers (Computer ke Prakar)

Size, speed aur capacity ke hisaab se computers alag-alag type ke hote hain.

Type	Size	Power	Use Case	Example
	Room-		Weather forecasting, scientific	PARAM,

Supercomputer	size	Sabse tez	simulation, nuclear research.	Fugaku
Mainframe	Large cabinet	Very High	Banks, airlines, large organizations (bulk data processing).	IBM zSeries
Mini Computer	Small cabinet	Medium	Mid-size companies ke servers. (Ab outdated, servers le aaye hain).	PDP-11
Micro Computer	Desktop/ Laptop	Low/Medium	Personal use. Ye hamara Personal Computer (PC) hai.	Desktop, Laptop
Workstation	Desktop	High	Engineering, animation, CAD. Micro se tez, strong processor, zyada RAM.	HP Z Series
Embedded	Very Small	Very Low	Kisi specific device ke andar computer.	Microwave, Washing Machine, ATM

1.6 Applications of Computers (Computer ke Upyog)

Aaj computer har jagah hai. Kuch major areas niche diye gaye hain:

Field	Application
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Education	Online classes, digital libraries, smartboards, research.
Healthcare	MRI, CT Scan, robotic surgery, patient records, telemedicine.
Banking	Online banking (NEFT/RTGS), ATM, trading, fraud detection.
Business	Accounting, data analysis, inventory management, video conferencing.
Entertainment	Movies (VFX), music, OTT (Netflix, Prime), video games.
Defence	Missile control, radar, communication, surveillance.
Government	Aadhar card, PAN card, weather forecasting, digital India services.
Science & Research	Space exploration (ISRO/NASA), DNA sequencing, LHC experiments.

Topic 1: Summary & Revision

Key Point	Detail
Computer Definition	Electronic machine, Input -> Process -> Output -> Storage.
Father of Computer	Charles Babbage (Analytical Engine).
First Programmer	Ada Lovelace.
First Electronic Computer	ENIAC (Vacuum Tubes par based).
5 Generations	Vacuum Tubes -> Transistors -> ICs -> VLSI (Microprocessor) -> ULSI (AI).
GIGO	Garbage In, Garbage Out. Accuracy ka base principle.

Speed Unit	Hz, MHz, GHz.
Types of Computers	Super, Mainframe, Mini, Micro (PC), Workstation, Embedded.
Diligence	Kabhi thakta nahi, lagatar same speed se kaam karta hai.

? MCQs - Apna Test khud kariye

1. Charles Babbage ko kya kaha jaata hai?

- A) Father of Internet
- B) Father of Computer
- C) First Programmer
- D) Father of Transistor

2. Third Generation computers mein main technology kya thi?

- A) Transistor
- B) Vacuum Tubes
- C) Integrated Circuits (ICs)
- D) Artificial Intelligence

3. "GIGO" ka full form kya hai?

- A) Get Input, Get Output
- B) Garbage In, Garbage Out
- C) Great Input, Great Output
- D) None of the above

4. Sabse powerful computer kaunsa hota hai jo weather forecasting ke liye use hota hai?

- A) Mainframe
- B) Supercomputer
- C) Laptop
- D) Embedded System

(Answers: 1-B, 2-C, 3-B, 4-B)

💬 Interview Questions

1. Q: What is a computer?

- **A:** (Simple, confident jawab) A computer is a programmable electronic

device that takes raw data as input, processes it according to a set of instructions (program), and produces meaningful information as output, which can also be stored for future use.

2. **Q:** Briefly explain the characteristics of a computer.

- **A:** Computer has Speed, Accuracy (GIGO principle), Diligence (no fatigue), huge Storage capacity, Versatility (can do multiple tasks), and Automation (works automatically once instructed). Its only limitation is that it has zero IQ and cannot think on its own.

3. **Q:** What are the five generations of computers?

- **A:** 1st Gen (Vacuum Tubes), 2nd Gen (Transistors), 3rd Gen (Integrated Circuits), 4th Gen (VLSI/Microprocessors), and 5th Gen (ULSI with Artificial Intelligence).

Practice Questions

1. Apne smartphone ko ek example maan kar batayein ki woh "Input-Process-Output" model ko kaise follow karta hai?
2. "Computer kabhi thakta nahi" aur "Computer ki apni koi akal nahi", dono statements ko real life se jod kar explain kariye.
3. Generations of Computers ki table ko ek baar khud se draw karein aur har generation ki do khaas baatein likhein.

Real-World Example for this Chapter: ATM Machine ek **Embedded**

Computer hai. Aap ATM card daalte hain (Input), machine aapke bank se verify karti hai aur amount check karti hai (Processing), fir aapko paise deti hai aur screen pe balance dikhati hai (Output). Ye 24x7 bina thake kaam karta hai — ye hai **Diligence**.

2. Number System (Sankhya Paddhati)

Wah! Chapter 1 khatam. Ab hum andar ke system mein jaate hain. Computer sirf 0 aur 1 samajhta hai, to ye numbers ke types ko samajhna bahut zaroori hai.

Naya Term: Number System Number system woh tareeqa hai jisse hum numbers ko likhte aur represent karte hain. Jaise hum apni daily life mein **Decimal** (0-9) use karte hain, computers **Binary** (0,1) use karte hain. Har system ka ek **Base** ya **Radix** hota hai.

2.1 Types of Number Systems

Number System	Digits Used	Base (Radix)	Where Used?
Decimal	0, 1, 2, 3, 4, 5, 6, 7, 8, 9	10	Human daily life (paisa, weight, distance).
Binary	0, 1	2	Core computer language (transistor ON=1, OFF=0).
Octal	0, 1, 2, 3, 4, 5, 6, 7	8	Old computer systems mein memory address ke liye (ab kam use hota hai).
Hexadecimal	0-9, A(10), B(11), C(12), D(13), E(14), F(15)	16	Memory address, color codes (#FFFFFF for white), MAC address, error codes.

Why base hota hai? Base ka matlab hai ki us system mein kitne unique symbols (digits) hain. Decimal mein 10 symbols hain (0 se 9), isliye base 10 hai. Binary mein sirf 2 symbols hain (0 aur 1), isliye base 2 hai.

2.2 Positional Value Samjho

Har digit ki ek position hoti hai, aur us position ki ek value hoti hai. Ye base ki power hoti hai.

$$3 \times 10^2 + 4 \times 10^1 + 5 \times 10^0 = 300 + 40 + 5 = 345$$

$$1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 8 + 0 + 2 + 1 = 11$$

Power yaad rakhne ki trick: Rightmost digit 0 power se shuru hoti hai, aur left mein jaate waqt power increase hoti jaati hai (...3,2,1,0).

2.3 Conversions (Rupantaran)

Bahut important topic hai interview aur exam ke liye.

Decimal to Binary (10 -> 2)

Rule: Number ko baar-baar 2 se divide karo, aur remainder (sheshphal) likhte jao. Jab tak quotient (bhagfal) 0 na ho jaye. Fir remainders ko niche se upar ki

jab jab tak quotient (bhagdar) 0 na ho jaye. Fir remainders ko niche se upar ki taraf padho.

Example: 25 in Decimal ko Binary mein badlo.

```
2 | 25 | Remainder
-----+-----
2 | 12 | 1      (25/2 = 12, remainder 1)
-----+-----
2 | 6  | 0
-----+-----
2 | 3  | 0
-----+-----
2 | 1  | 1
-----+-----
| 0  | 1      (Yahan rukna hai jab tak 0 na aaye)
```

Ab remainders ko **bottom to top** padho: **11001** Toh 25 (Decimal) = 11001 (Binary).

Binary to Decimal (2 -> 10)

Rule: Binary number ki har digit ko 2 ki increasing power (0,1,2...) se multiply karke jodo.

Example: 11001 ko Decimal mein badlo.

```
1      1      0      0      1
|      |      |      |      |
2^4    2^3    2^2    2^1    2^0
16     + 8     + 0     + 0     + 1 = 25
```

Total = 25.

Decimal to Octal & Hexadecimal

Rule: Bilkul binary jaise hi, bas 2 ki jagah 8 ya 16 se divide karo. **Hexadecimal**

Trick: Remainder agar 10 se 15 aaye to A-F se badlo.

Example: 25 ko Octal mein badlo. 25 / 8 -> Quotient=3, Remainder=1. 3 / 8 -> Quotient=0, Remainder=3. Bottom to top: **31**. Toh 25 (Dec) = 31 (Octal).

Example: 42 ko Hexadecimal mein badlo. 42 / 16 -> Q=2, Rem=10 (A) 2 / 16 -> Q=0, Rem=2 Bottom to top: **2A**. Toh 42 (Dec) = 2A (Hex).

Shortcut: Binary <-> Octal, Hexadecimal

Yeh bahut easy hai.

- **Binary to Octal:** Binary digits ko right se 3 ke group mein baanto, har group ka decimal equivalent likh do.

◦ **101011** -> Group right se: **101** **011** -> 5, 3 -> **(53)₈**

- **Binary to Hex:** Binary digits ko right se 4 ke group mein baanto, har group ka hex equivalent likho.

◦ **10101011** -> Group right se: **1010** **1011** -> A, B -> **(AB)₁₆**

2.4 Binary Arithmetic (Dviguniya Ganit)

Binary mein sirf jod, ghatana, guna, bhaag karna. Rules bahut simple hain.

- **Addition:**

- $0 + 0 = 0$
- $0 + 1 = 1$
- $1 + 0 = 1$
- $1 + 1 = 0$ (aur 1 carry ho jayega agle column mein, jaise decimal mein $5+5=10$ hota hai)

- **Subtraction (with borrowing):**

- $0 - 0 = 0$
- $1 - 0 = 1$
- $1 - 1 = 0$
- $0 - 1 = 1$ (Agle column se 1 borrow karke, jo ki 2 ban jaata hai. To $2-1=1$)

ASCII & Unicode (Abhi Introduction)

Computer sirf numbers samajhta hai. Toh humare letters, symbols ko number code dena pada. Isi ko **Character Encoding** kehte hain.

- **ASCII (American Standard Code for Information Interchange):** 7-bit code hai, total 128 characters store kar sakta hai (A-Z, a-z, 0-9, special symbols). e.g., 'A' = 65, 'a' = 97.
- **Unicode:** Duniya ki sabhi bhashaon (Hindi, Chinese, Arabic) ke liye ek universal encoding system. Ye 16-bit, 32-bit etc. hota hai. **UTF-8** iska sabse popular format hai.

Topic 2: Summary & Revision

Topic	Key Point
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Term	Key Point
Number Systems	Decimal(10), Binary(2), Octal(8), Hex(16).
Base/Radix	Unique digits count in system.
Dec -> Binary	Divide by 2, note remainders, read bottom-up.
Bin -> Dec	Multiply digits by powers of 2, sum up.
Shortcut Groups	Octal = 3-bit group, Hex = 4-bit group.
ASCII	7-bit, 128 chars. 'A'=65.
Unicode	Universal, multi-byte. UTF-8 most common.

? MCQs

1. **Decimal number 15 ka binary equivalent kya hai?**

- A) 1100
- B) 1111
- C) 1000
- D) 1110

2. **Hexadecimal system mein 'D' kis value ko represent karta hai?**

- A) 12
- B) 14
- C) 13
- D) 15

3. **Binary addition mein 1 + 1 ka result kya hota hai?**

- A) 2
- B) 11
- C) 0 (Carry 1)
- D) 1 (Carry 0)

4. **"Computer sirf 0 aur 1 samajhta hai." Ye kaunsa number system hai?**

- A) Decimal
- B) Hexadecimal
- C) Binary
- D) Octal

(Answers: 1-B, 2-C, 3-C, 4-C)

Interview Questions

1. **Q:** Convert $(45)_{10}$ into Binary and Hex.
 - **A:** $45 / 2 \rightarrow$ Binary is **101101** . $45 / 16 \rightarrow$ Hex is **2D** . (Process explain karna step-by-step).
 2. **Q:** Why do computers use binary instead of decimal?
 - **A:** Computers electronic circuits (transistors) se bane hain, jinki do states hoti hain — ON (1) aur OFF (0). Binary system inhi do states ko perfectly represent karta hai, jo bahut reliable hai. Decimal ke liye 10 states chahiye hote, jo electronics mein manage karna mushkil hai.
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Practice Questions

1. Apne birth year (jaise 2002) ko Decimal se Binary, Octal aur Hexadecimal mein convert kariye.
 2. Binary number **10101010** ko shortcut se Octal aur Hexadecimal mein badle bina usse Decimal mein convert kiye.
 3. Ek program sochiye jo kisi bhi value ko check karta ho. Agar woh **0** hai to OFF kare aur **1** hai to ON kare. Binary ki is simplicity par 2 line likhiye.
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Shivani, ye raha humara starting ka solid base. Is safar mein aap dheere-dheere ye sab terms aur concepts use karna seekh jayengi. Main chahta hoon ki aap isse ek baar zaroor padhein, MCQs solve karein, aur jo bhi doubt aaye poochhiye.

Agla part hum shuru karenge jab aap ready hongi. **Data Representation** aur **Hardware** mein hum computer ki internal working aur memory ko explore karenge — jaise bit, byte, RAM, ROM, CPU wagera.

Aapko is topic ki summary aur practice material kaisa laga? Next part ke liye taiyaar hain?